FORESIGHT - Work Distribution

Use together with **FORESIGHT_Build_Spec.md**. Everyone should read the full specification once; the sections below are the primary areas each person should implement and maintain.

Shared priority: finish Tier A end-to-end first. Work in parallel only after the common contracts, schemas, time rules and feature dimensions are frozen. Optional Tier B/C work should not delay the core.

Person	Primary area	Work from Build Spec
Aarav Mehta	Core World Model + Integration	**Primary sections:** 13-15, 25-26, 35; Appendix H, I, S, T. **Build:** directed GAT, per-entity GRU memory, global LSTM, transition prior, rollout, prediction heads, model interfaces, integration of all modules, config/release consistency. **Repo areas:** `models/`, core rollout, integration, configs/release. **Done when:** model shapes/contracts pass; synthetic vertical slice runs through real interfaces; all merged modules remain compatible with the Build Spec.
Yuthika Singh	Datasets + Labels + Training	**Primary sections:** 6, 11, 15, 29; Appendix D, I, M, T. **Build:** dataset registry, qualified attack episodes, chronological splits, target/risk-set construction, target masks, training clips/loaders, training pipeline, checkpoints and training manifests. **Repo areas:** `data/`, `labels/`, `training/`. **Done when:** labels have traceable provenance; no leakage across splits; training can reproduce from a manifest.
Srishtee Varule	PCAP/CSV + Features + Graphs	**Primary sections:** 7-10; Appendix D-G, Q, R. **Build:** safe PCAP/CSV ingestion, canonical events, prefix selection, flow segmentation, packet/flow features, validity masks, identity/scope handling, transforms and directed graph construction. **Repo areas:** `ingest/`, `identity/`, `features/`, `graphs/`. **Done when:** exact 88/22/48 contracts pass; suffix changes cannot alter earlier features; graph/identity tests pass.
Zeshawn Martis	Trust + SENTINEL + Evaluation	**Primary sections:** 16-19, 29; Appendix J, K, M, Q, R. **Build:** calibration, abstention/trust contract, SENTINEL novelty, baselines, metrics, held-out-family tests, ablations, robustness tests, explanation fidelity and evaluation reports. **Repo areas:** `calibration/`, `sentinel/`, `evaluation/`, `explain/`. **Done when:** metrics match hand calculations; unknown is not converted to benign; every headline claim has a reproducible comparison.
Dayaan Jain	Backend + Runtime + Evidence	**Primary sections:** 25, 27-28; Appendix E, L, Q, S, T. **Build:** FastAPI runtime, sessions/jobs, immutable object storage, ForecastRecord persistence, forecast sealing, ledger, reveal/evaluator separation, health/readiness, artifact verification and error/status handling. **Repo areas:** `runtime/`, `storage/`, `api/`. **Done when:** forecast is committed before reveal; future truth is inaccessible to inference; restart/idempotency/error tests pass.
Saahil Grover	Analyst Console + Demo + E2E	**Primary sections:** 19, 25, 30; Appendix K, L, R, T. **Build:** local analyst console, upload/session screens, network graph, risk/ETA timeline, SENTINEL/trust display, evidence panels, reveal workflow, evaluation/export screens, UI failure states and end-to-end tests. **Repo areas:** `apps/console/`, `tests/e2e/`, demo/release scripts. **Done when:** UI never invents/recomputes scientific results; mocked fixtures are replaced by real API output in release; full demo is reproducible.

How to Work Together in GitHub

The goal is to keep interfaces stable while allowing everyone to build in parallel.

1	**Freeze shared contracts first.** Push the Build Spec, schemas, config structure, time rules, feature registry and golden fixture before deep module work.
2	**Use feature branches.** Recommended: `feat/world-model`, `feat/data-labels`, `feat/ingest-graphs`, `feat/trust-evaluation`, `feat/runtime-api`, `feat/console`.
3	**Develop against typed fixtures.** If another module is not ready, use the synthetic/golden contract fixture rather than inventing a different interface.
4	**Every module includes its own tests.** Unit/contract tests stay with the module; E2E tests validate the full pipeline.
5	**Merge through pull requests.** Any change to time semantics, feature order, target meaning, model dimensions, forecast schema or evaluation policy must be discussed before merge.
6	**Integrate early.** First target is one tiny legal vertical slice: input -> features -> graph -> model -> forecast -> seal -> API -> UI -> reveal/evaluate.
7	**Tier A before Tier B/C.** Optional paths, active perception, intervention and continual learning remain disabled until the core evidence bundle works.

Integration Map

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Srishtee -> legal events / features / graphs --\
Yuthika -> labels / clips / training -------+-> Aarav -> model / rollout
Zeshawn -> calibration / trust / eval ------+-----------> Dayaan -> API / ledger -> Saahil ->
console / demo
```

Definition of Done for Each Part

A part is complete only when its inputs/outputs are typed and documented, normal/empty/invalid cases are tested, time/scope boundaries are respected, artifacts are reproducible, and it works in the integrated selected profile. Interfaces existing on paper do not count as implemented functionality.

First shared milestone: everyone should be able to clone the repository and run one small end-to-end fixture through the real interfaces. Accuracy can improve later; integration correctness should come first.